

Variational Autoencoders — Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Part A: Conceptual Questions

A1. Autoencoders vs. Variational Autoencoders

Question. A standard (deterministic) autoencoder and a VAE both use an encoder–decoder architecture. Identify two fundamental differences between them from a probabilistic perspective. In particular, explain what the encoder outputs in each case and why a standard autoencoder cannot be used to generate new samples by sampling from the latent space.

A2. The ELBO: Two Forms

Question. The Evidence Lower Bound can be written in two equivalent forms:

$$\begin{aligned}\mathcal{L}_{\theta,\phi}(x) &= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x, z) - \log q_\phi(z|x)], \\ \mathcal{L}_{\theta,\phi}(x) &= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - D_{\text{KL}}(q_\phi(z|x) \parallel p(z)).\end{aligned}$$

Show algebraically that these two forms are equivalent. Then interpret each term in Form 2 in the context of a VAE trained on sensor signals.

A3. Why the Marginal Likelihood is Intractable

Question. The VAE avoids computing the marginal likelihood $p_\theta(x) = \int p(z) p_\theta(x|z) dz$ directly.

- (a) Explain why this integral is intractable when the decoder $p_\theta(x|z)$ is a neural network.
- (b) The lecture gives two analytical examples: (i) discrete z , continuous x ; and (ii) continuous z , discrete x . For each, describe what “exact computation” would require and why it is infeasible.

A4. The Reparameterization Trick

Question. During training, the VAE must differentiate through a stochastic sampling operation $z \sim q_\phi(z|x)$.

- (a) Explain why naively differentiating $\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[f(z)]$ by moving the gradient inside the integral fails to yield a useful Monte Carlo estimator.
- (b) Describe the reparameterization trick for a Gaussian $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi(x)^2))$. Write the explicit form of z as a deterministic function of ϕ and a noise variable ϵ .
- (c) State the two gradient paths that arise from the reparameterized form and explain why this estimator has lower variance than the score-function estimator.

A5. Posterior Collapse

Question. Posterior collapse is a failure mode specific to VAEs (and not plain autoencoders).

- (a) Describe the symptom of posterior collapse in terms of the encoder output and the KL term during training.
- (b) Explain the mechanism that causes it: why does the training dynamics “prefer” this degenerate solution early in training?
- (c) Describe two remedies — β -VAE / KL annealing and Free Bits — and explain how each prevents the collapse.

A6. Deterministic vs. Probabilistic Inference After Training

Question. After a VAE is trained, one can use the encoder in two modes: deterministic ($z^* = \mu_\phi(x^*)$) or probabilistic ($z^* = \mu_\phi(x^*) + \sigma_\phi(x^*) \odot \epsilon$). For each of the following downstream tasks, state which mode is more appropriate and justify your choice:

- (a) Extracting fixed-length feature vectors for a downstream classifier,
- (b) Estimating uncertainty in the reconstruction of a noisy sensor reading,
- (c) Generating diverse synthetic training data for data augmentation.

A7. Score-Function Estimator for Discrete Latent Variables

Question. When the latent variable z is discrete, the reparameterization trick cannot be applied.

- (a) State the score-function (REINFORCE) estimator for $\nabla_\phi \mathbb{E}_{q(z|\phi)}[\log p_\theta(x|z)]$. Identify which term acts as the “reward signal.”
- (b) Explain why this estimator is unbiased but has high variance.

Part B: Mathematical Derivations

B1. Deriving the ELBO from the Marginal Log-Likelihood

Problem. Starting from $\log p_\theta(x)$, introduce the variational distribution $q_\phi(z|x)$ via importance weighting and apply Jensen's inequality to derive the ELBO:

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\text{KL}}(q_\phi(z|x) \| p(z)).$$

Identify the gap between $\log p_\theta(x)$ and the ELBO and state what it equals.

B2. Analytic KL for Diagonal Gaussians

Problem. Let $q_\phi(z|x) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$ and $p(z) = \mathcal{N}(0, I)$, with $z \in \mathbb{R}^d$.

- (a) Derive the closed-form expression for the scalar case $d = 1$:

$$D_{\text{KL}}(q\|p) = \frac{1}{2}(\sigma^2 + \mu^2 - 1 - \log \sigma^2).$$

- (b) Extend to the d -dimensional diagonal case. Write the final formula.
(c) Verify the two sanity checks: (i) $\mu = 0, \sigma = 1$ gives $D_{\text{KL}} = 0$; (ii) $\mu = 1, \sigma = 1$ gives $D_{\text{KL}} = \frac{1}{2}$.

B3. Gaussian VAE Loss Function

Problem. Assume the decoder is Gaussian: $p_\theta(x|z) = \mathcal{N}(x; \hat{x}_\theta(z), \sigma^2 I)$.

- (a) Show that $\mathbb{E}_{q_\phi}[\log p_\theta(x|z)]$ reduces (up to constants) to a scaled mean squared error:

$$\mathbb{E}_{q_\phi}[\log p_\theta(x|z)] = -\frac{1}{2\sigma^2} \mathbb{E}_{q_\phi}[\|x - \hat{x}_\theta(z)\|^2] + \text{const.}$$

- (b) Write out the complete practical loss minimized in the VAE (sign-flipped ELBO), including the β -VAE generalization. Identify what σ^2 and β each control in the reconstruction/regularization trade-off.

B4. Reparameterization: Gradient Computation

Problem. Let $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, and let $f(z) = \log p_\theta(x|z)$.

- (a) Write the Monte Carlo estimator for $\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[f(z)]$ using S samples $\epsilon^{(1)}, \dots, \epsilon^{(S)}$.
(b) Using the chain rule, identify the two gradient paths: $\partial z / \partial \mu_\phi$ and $\partial z / \partial \sigma_\phi$, and write the explicit gradient expression.

B5. VAE for Time Series: Window ELBO

Problem. A window-based VAE processes a length- T time series window $x \in \mathbb{R}^T$ with decoder $p_\theta(x|z) = \mathcal{N}(\mu_\theta(z), \sigma^2 I_T)$ and prior $p(z) = \mathcal{N}(0, I_d)$.

- (a) Write the per-window ELBO $\mathcal{L}(x; \theta, \phi)$ explicitly, showing both the reconstruction term (as MSE) and the KL term (in closed form).
- (b) An anomaly score for window x is defined as the negative ELBO: $A(x) = -\mathcal{L}(x; \theta, \phi)$. Decompose $A(x)$ into a reconstruction component and a KL component, and explain intuitively why a healthy window and an anomalous window produce different scores.

Part C: Parametric Analysis

C1. Effect of the Latent Dimension d

Question. Consider a VAE trained on respiratory sensor windows. How does increasing the latent dimension d affect: (a) the model's capacity to represent complex breathing patterns, (b) the risk of posterior collapse, and (c) the interpretability of the latent space? What practical heuristic does the lecture recommend for choosing d ?

C2. Effect of β in the β -VAE

Question. The β -VAE loss is:

$$\mathcal{L}_\beta = \mathbb{E}_q[\log p_\theta(x|z)] - \beta D_{\text{KL}}(q_\phi(z|x) \| p(z)).$$

Describe qualitatively what happens to the learned representations and reconstruction quality as: (a) $\beta \rightarrow 0$, (b) $\beta = 1$ (standard VAE), (c) $\beta \gg 1$. What does a very large β encourage in the latent space?

C3. KL Annealing Schedule

Question. In the lecture, KL annealing starts with $\beta \approx 0$ and linearly increases to $\beta = 1$ over training. Explain: (a) why starting with $\beta = 0$ helps the decoder learn first, and (b) what risk arises if the annealing schedule is too fast (i.e., β reaches 1 very quickly). How does this differ from the Free Bits approach in terms of what each method controls?

C4. Gaussian vs. Bernoulli Decoder

Question. The lecture recommends using a Gaussian decoder (MSE loss) for continuous signals and a Bernoulli decoder (binary cross-entropy loss) for binary data.

- (a) Write the per-sample log-likelihood $\log p_\theta(x|z)$ for the Bernoulli decoder with output probability $\pi_\theta(z) \in (0, 1)^D$.

Part D: Application and Notebook Questions

D1. Engineering Application – Breathing Anomaly Detection

You are given the `VAE_Breathing_Anomaly` notebook, which trains a window-based VAE on respiratory sensor data to detect abnormal breathing events.

- (a) The notebook computes an anomaly score for each test window. Describe the two components of this score (reconstruction error and KL term) and explain why both contribute to anomaly detection: what does each component capture that the other does not?
- (b) During training, the notebook tracks reconstruction loss and KL divergence separately. Sketch the expected trajectories of both quantities during healthy training. Then describe what “posterior collapse” looks like on these two training curves.
- (c) The notebook uses a latent dimension of $d = 8$. Propose an experiment to select an appropriate d for this task. What metric would you use to compare models with $d \in \{2, 4, 8, 16, 32\}$ and which data split would you use?
- (d) Modify the notebook to implement KL annealing: let β grow linearly from 0 to 1 over the first 50 epochs. Compare the final ELBO, reconstruction loss, and anomaly detection AUC against the baseline (fixed $\beta = 1$). What do you observe?

D2. Latent Space Visualization and Interpretation

- (a) After training, the notebook projects test windows into the latent space using $z^* = \mu_\phi(x^*)$ and plots them in 2D (using PCA or t-SNE if $d > 2$). What geometric property of the latent space would you expect for a well-trained VAE (as opposed to a deterministic autoencoder)? Explain why this property is enforced by the KL term.
- (b) Normal and anomalous breathing windows are encoded separately. Describe where you would expect anomalous windows to appear in the latent space relative to normal windows, and relate this to the KL anomaly score component.
- (c) Propose a threshold-selection procedure for the anomaly score $A(x)$ that uses only the normal training data. Justify your choice of threshold in terms of the latent space geometry and the prior $p(z) = \mathcal{N}(0, I)$.

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